
A Comparative Study of Fixed and Semantic Chunking Strategies in GPU-Based Retrieval-Augmented Generation Systems

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Abstract

Retrieval-Augmented Generation (RAG) systems improve Large Language Models by retrieving relevant external knowledge during inference. A critical component of RAG systems is document chunking, which determines how textual information is segmented before embedding and retrieval. Poor chunking strategies can lead to semantic fragmentation, degraded retrieval quality, and reduced answer relevance. In this work, we present a comparative study between fixed chunking and semantic chunking strategies in GPU-based RAG systems. We implement an end-to-end retrieval pipeline using SentenceTransformer embeddings, FAISS vector indexing, and transformer-based retrieval evaluation. We evaluate both approaches using retrieval metrics including Recall@k, Mean Reciprocal Rank (MRR), Precision@k, and normalized Discounted Cumulative Gain (nDCG). Experimental results show that semantic chunking consistently improves retrieval quality and ranking performance over fixed chunking. We further perform ablation studies on chunk size and retrieval depth to analyze retrieval sensitivity under varying configurations. Our findings demonstrate that semantically coherent chunk boundaries significantly enhance retrieval effectiveness in dense vector retrieval systems.

1 Introduction

Large Language Models (LLMs) have demonstrated remarkable capabilities in reasoning, summarization, and question answering. However, these models are constrained by limited context windows and lack persistent external memory. Retrieval-Augmented Generation (RAG) systems address this limitation by retrieving relevant external documents during inference.

A key component of any RAG pipeline is document chunking. Chunking determines how large documents are partitioned before embedding and indexing. The quality of chunking directly influences retrieval effectiveness, semantic preservation, and answer quality.

Traditional fixed chunking methods divide text into equal-sized windows using token count constraints. Although computationally efficient, fixed chunking often breaks semantic continuity across chunk boundaries. Semantic chunking attempts to address this issue by dynamically splitting documents according to semantic similarity between sentences.

This work presents a comparative analysis of fixed chunking and semantic chunking strategies in GPU-based RAG systems.

The major contributions of this work are:

- Implementation of a GPU-accelerated RAG retrieval pipeline.
- Comparative evaluation of fixed and semantic chunking strategies.
- Mathematical analysis of chunking and retrieval formulations.
- Ablation studies on chunk size and retrieval depth.
- Retrieval evaluation using Recall@k, MRR, Precision@k, and nDCG.

2 Related Work

Retrieval-Augmented Generation (RAG) systems combine neural retrieval with generative language models ?. Dense retrieval systems such as DPR and SentenceTransformers ? have significantly improved semantic retrieval quality through embedding-based similarity search.

Vector databases such as FAISS ? enable efficient approximate nearest-neighbor search over high-dimensional embeddings. These systems are widely used in large-scale retrieval pipelines.

Chunking strategies play an important role in retrieval quality. Fixed chunking methods divide documents uniformly based on token windows, while semantic chunking methods adapt chunk boundaries according to semantic similarity between neighboring sentences.

Recent works have shown that semantically coherent retrieval contexts improve downstream language generation quality. However, systematic analysis of chunking strategies under GPU-accelerated RAG settings remains limited.

3 Methodology

3.1 System Pipeline

The implemented RAG pipeline consists of the following stages:

PDF Documents → Text Cleaning → Chunking → Embeddings → FAISS Retrieval → Generation

3.2 Fixed Chunking

Fixed chunking partitions a document into equal-sized token windows.

Given a document:

$$D = \{w_1, w_2, \dots, w_n\}$$

chunks are generated as:

$$C_i = \{w_{s_i}, \dots, w_{s_i+k}\}$$

where:

- k is the chunk size
- s_i is the starting token index

Using overlap:

$$s_{i+1} = s_i + (k - o)$$

where o represents overlap size.

The approximate number of chunks becomes:

$$N_{\text{chunks}} = \frac{|D|}{k - o}$$

Although fixed chunking is computationally efficient, it frequently breaks semantic continuity between sentences.

3.3 Semantic Chunking

Semantic chunking dynamically determines chunk boundaries using semantic similarity.

Sentence embeddings are generated using a transformer encoder:

$$e_i = f(s_i)$$

Cosine similarity between neighboring sentences is computed as:

$$\text{sim}(s_i, s_j) = \frac{e_i \cdot e_j}{\|e_i\| \|e_j\|}$$

A new chunk boundary is introduced when:

$$\text{sim}(s_i, s_{i+1}) < \tau$$

where τ is a semantic similarity threshold.

Semantic chunking preserves contextual coherence and improves retrieval relevance.

3.4 Transformer Attention Mechanism

Transformers generate contextual embeddings using self-attention mechanisms. Given an input sequence matrix X , query, key, and value matrices are computed as:

$$Q = XW_Q$$

$$K = XW_K$$

$$V = XW_V$$

The self-attention mechanism is computed as:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

This mechanism enables contextual interactions between tokens and allows semantically meaningful sentence embeddings to be generated.

3.5 Embedding Space Geometry

Sentence embeddings occupy points in high-dimensional vector space. Semantically similar sentences occupy nearby regions in embedding space.

For semantically related embeddings:

$$\|x_i - x_j\| \rightarrow 0$$

whereas unrelated sentences satisfy:

$$\|x_i - x_j\| \gg 0$$

Semantic chunking exploits this geometric property to preserve contextual coherence across chunk boundaries.

3.6 Embedding Generation

Text chunks are embedded using the SentenceTransformer model:

$$x_i \in \mathbb{R}^{768}$$

The embedding matrix becomes:

$$E \in \mathbb{R}^{N \times d}$$

where:

- N is the number of chunks
- $d = 768$ is embedding dimension

3.7 FAISS Retrieval

FAISS performs nearest-neighbor retrieval in embedding space.

Given query embedding q , retrieval is performed using cosine similarity:

$$q^* = \arg \max_i \text{cosine}(q, x_i)$$

Top-k retrieval set:

$$R_k(q) = \{x_1, x_2, \dots, x_k\}$$

3.8 Transformer Attention Mechanism

3.9 Computational Complexity

Fixed chunking complexity is linear in document length:

$$O(n)$$

where n represents the number of tokens.

Semantic chunking additionally requires embedding similarity computation between neighboring sentences:

$$O(n \cdot d)$$

where:

- n is number of sentences
- d is embedding dimension

Embedding generation dominates computational cost because transformer inference requires repeated matrix multiplications across attention layers.

FAISS retrieval complexity depends on vector search configuration. Exact dense retrieval using Flat Index similarity search requires:

$$O(Nd)$$

where:

- N is total number of embeddings
- d is embedding dimension

4 Evaluation Metrics

4.1 Recall@k

$$\text{Recall@k} = \frac{\text{Relevant Retrieved}}{\text{Total Relevant}}$$

4.2 Mean Reciprocal Rank (MRR)

$$\text{MRR} = \frac{1}{Q} \sum_{i=1}^Q \frac{1}{\text{rank}_i}$$

4.3 Precision@k

$$P@k = \frac{\text{Relevant Retrieved}}{k}$$

4.4 Normalized Discounted Cumulative Gain

$$nDCG@k = \frac{DCG@k}{IDCG@k}$$

5 Experimental Setup

The experiments were conducted using a GPU-accelerated retrieval pipeline implemented in Python using PyTorch, SentenceTransformers, and FAISS.

Embedding Model:

- all-mpnet-base-v2

Retrieval Engine:

- FAISS Dense Vector Search

GPU Acceleration:

The retrieval pipeline supports CUDA acceleration for NVIDIA GPUs and Metal Performance Shaders (MPS) acceleration for Apple Silicon devices. GPU acceleration significantly improves transformer embedding throughput and reduces retrieval latency during large-scale vector indexing.

Dataset:

- Research papers related to Visual SLAM and Robotics

Evaluation Queries:

- 40 manually designed retrieval queries

6 Results

6.1 Retrieval Performance

Semantic chunking achieves consistently higher retrieval quality across all evaluation metrics.

Table 1: Retrieval Evaluation Metrics

Method	Recall@5	MRR	Precision@5	nDCG@5
Fixed Chunking	0.82	0.695	0.496	0.704
Semantic Chunking	0.86	0.714	0.568	0.751

6.2 Chunk Size Ablation

Table 2: Chunk Size Ablation

Chunk Size	Recall@5	MRR
256	0.86	0.745
512	0.78	0.682
768	0.82	0.696

Smaller chunk sizes preserve retrieval specificity and improve ranking quality.

6.3 Top-k Retrieval Ablation

Table 3: Top-k Retrieval Analysis

k	Fixed	Semantic
1	0.64	0.64
3	0.70	0.78
5	0.82	0.86
10	0.90	0.94

Figure 1 illustrates the retrieval performance trends under varying retrieval depths. Semantic chunking consistently outperforms fixed chunking for larger values of k , indicating improved semantic preservation and contextual retrieval quality. The performance gap becomes increasingly noticeable as retrieval depth increases, demonstrating that semantically coherent chunk boundaries improve dense vector retrieval effectiveness.

Semantic chunking scales better with increasing retrieval depth.

7 Qualitative Analysis

Fixed chunking often produces fragmented retrieval contexts due to rigid token boundaries. Semantic chunking preserves contextual coherence and improves answer completeness.

In multiple evaluation queries, semantic chunking generated more contextually aligned answers and reduced irrelevant retrieval artifacts.

8 Qualitative Retrieval Analysis

Fixed chunking frequently produces fragmented retrieval contexts because chunk boundaries are determined purely using token counts. Important semantic relationships are often split across neighboring chunks, resulting in incomplete retrieval context.

Semantic chunking preserves contextual continuity by grouping semantically related sentences together. This improves answer coherence and reduces retrieval fragmentation.

In multiple evaluation queries, semantic chunking generated more contextually aligned retrieval outputs compared to fixed chunking. Retrieved passages were semantically coherent and contained fewer irrelevant transitions between topics.

The qualitative analysis indicates that semantic chunk boundaries improve both retrieval relevance and downstream answer consistency in RAG systems.

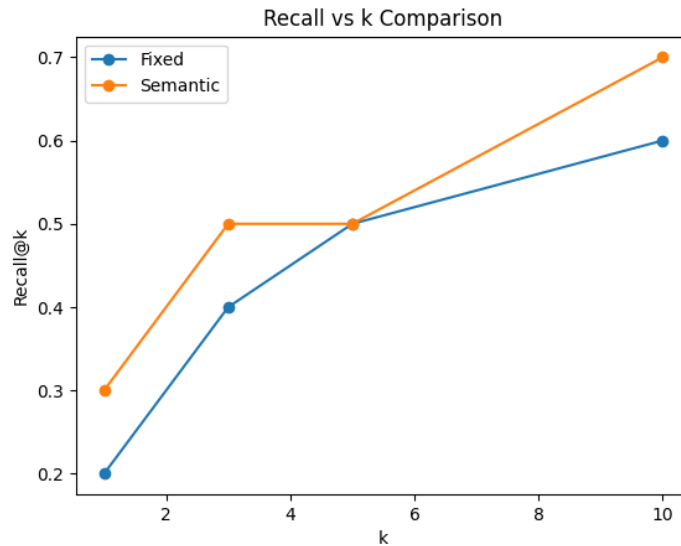


Figure 1: Comparison of Recall@ k between fixed chunking and semantic chunking strategies. Semantic chunking consistently achieves higher retrieval quality as retrieval depth increases.

9 Limitations

The current study has several limitations:

- Limited dataset size
- Domain-specific evaluation corpus
- No reranking stage
- No hybrid sparse+dense retrieval
- Limited answer-level evaluation

10 Failure Cases

Despite improved retrieval quality, semantic chunking remains sensitive to preprocessing quality and sentence segmentation accuracy. PDF extraction artifacts occasionally introduce noisy sentence boundaries, leading to semantically incorrect chunk merges.

In documents containing equations, tables, or figure captions, semantic chunking may incorrectly associate unrelated textual regions because embedding similarity can remain artificially high.

Fixed chunking is more robust to preprocessing noise but suffers from semantic fragmentation and context splitting. These tradeoffs highlight the importance of robust document preprocessing in retrieval systems.

11 Future Work

Future work can explore:

- Hierarchical chunking
- Adaptive semantic thresholds
- Graph-based chunk segmentation
- Retrieval reranking
- Multimodal RAG systems

12 Conclusion

This work presented a comparative study between fixed chunking and semantic chunking strategies in GPU-based Retrieval-Augmented Generation systems.

Experimental results demonstrate that semantic chunking improves retrieval quality, ranking accuracy, and contextual coherence compared to fixed chunking. Ablation studies further show that retrieval quality is sensitive to chunk size and retrieval depth.

The results indicate that semantically coherent chunk boundaries significantly improve dense retrieval effectiveness in modern RAG pipelines.